Tokens-per-Parameter Coverage at Scale: A Larger-Model Validation of Scaling-Law Conditioning

Large-scale validation study
Companion analysis to *"Tokens-per-Parameter Coverage Is Critical for Robust LLM Scaling Law Extrapolation"*

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Abstract

We validate the estimation-conditioning theory of the tokens-per-parameter (TPP) coverage study on an independent, substantially larger training grid: 461 language-model pretraining runs spanning non-embedding parameter counts from 5.7×10^4 to 3.5×10^8 (total parameters up to 5×10^8), token budgets from 10^8 to 10^{11}, trained on C4 over a TPP band of $[20, 200]$. Reproducing the paper's full estimation pipeline (four-law fits, Gauss-Newton conditioning, the closed-form TPP-diversity threshold, and the holdout regime classification), we find three results. First, the near-equality of the parameter- and data-scaling exponents *persists* at this larger scale: the exponent gap is $\varepsilon = 0.085$ for Chinchilla, 0.043 for Kaplan, and 0.021 for Droppo-Elibol, matching the small-scale values reported in the main text, so the ill-conditioning hazard is real rather than an artifact of tiny models. Second, the diverse-TPP two-dimensional design is *well-conditioned* (Regime B: the TPP-diversity statistic V_K exceeds the threshold τ_K for target condition numbers up to 10^6; the (A, B)-block condition number is $\kappa_{A,B} = 86$). Third, a single-TPP collinear ray drawn from the *same* runs is *ill-conditioned* (Regime A), with $\kappa_{A,B}$ up to 1.9×10^7, a five-order-of-magnitude penalty consistent with the predicted $\kappa = \Theta(\varepsilon^{-2})$ blow-up. A budget-matched holdout comparison favors the non-collinear design (median RMSE 0.073 vs 0.085; win rate 0.67). Rerunning the paper's estimation pipeline, every formal statement reproduces on this grid: the conditioning law $\kappa_{A,B} \propto \varepsilon^{-1.98}$ and the CI inflation $\propto \varepsilon^{-1.01}$ match the predicted ε^{-2} and ε^{-1}, the closed-form diversity threshold and holdout regimes hold, and the isoFLOP invariance is exact. These findings confirm, at up to 6.5× larger scale than the main experiments (5 to 77M parameters), both the hazard the theory identifies and the design recipe it prescribes.

1 Introduction and motivation

The companion paper argues that fitting neural scaling laws from runs performed at a single tokens-per-parameter (TPP) ratio k (the compute-optimal $D = kN$ convention) induces an inherent ill-conditioning in the Gauss-Newton least-squares problem: because the parameter-scaling and data-scaling exponents are empirically near-equal, the two scale-coefficient columns of the Jacobian become near-proportional, the condition number grows as $\Theta(\varepsilon^{-2})$ in the exponent gap ε, and the scale coefficients become practically unidentifiable. The paper proves this for four scaling-law formalisms, derives a closed-form TPP-diversity threshold that is necessary and sufficient for well-conditioned estimation, and shows empirically that non-collinear (NC) designs beat collinear (CO) ones on held-out data.

The main-text experiments use models of 5 to 77M parameters. A natural question is whether the phenomenon, and the prescribed remedy, survive at larger scale. This report answers that question directly. We take an *independent* grid of language-model pretraining runs, one that was designed as a genuine two-dimensional (N, D) sweep with a wide TPP band, and we run the paper's entire analysis on it: we fit all four laws, measure the exponent gaps, compute the Gauss-Newton conditioning ($\kappa(J^\top J)$ and the (A, B)-block $\kappa_{A,B}$), evaluate the TPP-diversity statistic V_K against its threshold τ_K, classify the resulting regime, and compare collinear against non-collinear designs on off-manifold holdouts.

The headline is a clean two-sided confirmation. The *hazard* is real at scale: the exponent near-equality that drives the degeneracy holds just as tightly for these larger models as for the small ones. The *remedy* works at scale: the diverse-TPP two-dimensional design is well-conditioned (Regime B), whereas a single-TPP ray carved from the identical runs is ill-conditioned (Regime A) with a six-order-of-magnitude condition-number penalty.

2 Background: the conditioning theory

We recall the objects we reproduce. The Chinchilla law is

$$L(N, D) = A\,N^{-\alpha} + B\,D^{-\beta} + E, \tag{1}$$

with the analogous Kaplan and Droppo-Elibol forms. Fitting is by nonlinear least squares with residuals $r_i = L_i - \hat{L}(N_i, D_i; \boldsymbol{\theta})$ and Jacobian $J_{ij} = \partial r_i / \partial \theta_j$; we write $\kappa(M) = \lambda_{\max}(M)/\lambda_{\min}(M)$ and let $\kappa_{A,B} = \kappa((J^\top J)_{A,B})$ be the condition number of the (A, B) principal submatrix (the two scale coefficients).

Exponent gap. The law-specific gap is $\varepsilon = |\alpha - \beta|$ for Chinchilla, $|\alpha_D - \alpha_N|$ for Kaplan, and $|\gamma_N - \gamma_D|$ with $\gamma_\bullet = \alpha_\bullet/\alpha$ for Droppo-Elibol.

Conditioning (Prop. 1). On a collinear ray $D = kN$ the two scale-coefficient columns satisfy $\mathbf{j}_B = c\,\mathbf{j}_A + \boldsymbol{\delta}$ with $\|\boldsymbol{\delta}\| = O(\varepsilon)$, so as $\varepsilon \to 0$, $\lambda_{\min}(J^\top J) = \Theta(1)$, $\lambda_{\min}(J^\top J) = \Theta(\varepsilon^2)$, and

$$\kappa(J^\top J) = \Theta(\varepsilon^{-2}). \tag{2}$$

The confidence interval on A inflates by $\Theta(\varepsilon^{-1})$ relative to the identified combination $\psi = A + Bk^{-\alpha}$ (Cor. 1).

TPP diversity and regimes (Prop. 2, Thm. 1). For K ordered training ratios $k_1 \le \cdots \le k_K$, and to leading order as $\varepsilon \to 0$, $\kappa(J^\top J) \le \kappa_{\text{target}}$ if and only if

$$V_K \;\coloneqq\; \frac{1}{K}\sum_\ell k_\ell^{-2\beta_{\text{eff}}} - \Big(\frac{1}{K}\sum_\ell k_\ell^{-\beta_{\text{eff}}}\Big)^2 \;\;\ge\;\; \tau_K \;\coloneqq\; \frac{\big(K + \sum_\ell k_\ell^{-2\beta_{\text{eff}}}\big)^2}{K^2\,\kappa_{\text{target}}}, \tag{3}$$

where β_{eff} is the data-side exponent. Regime A ($V_K < \tau_K$, ill-conditioned, including every $K = 1$ ray) gives $\mathbb{E}[\text{RMSE}^{\text{NC}}] < \mathbb{E}[\text{RMSE}^{\text{CO}}]$; Regime B ($V_K \ge \tau_K$, well-conditioned, requires $K \ge 2$) gives $\mathbb{E}[\text{RMSE}^{\text{CO}}]/\mathbb{E}[\text{RMSE}^{\text{NC}}] = \Theta(1)$.

Table 1: Four-law fits on the full (29-cell) two-dimensional envelope. ε is the law-specific exponent gap; smaller means more collinearity-prone. Fit quality is high throughout.

Law	exponents	ε	R^2	RMSE	notes
Chinchilla	α=0.123, β=0.038	0.085	0.9985	0.032	$E \to 0$ at bound
Kaplan	α_N=0.078, α_D=0.121	0.043	0.9722	0.139	N_c, D_c at bound
Droppo-Elibol	γ_N=0.244, γ_D=0.264	0.021	0.9985	0.033	L_∞=2.02

3 Experimental setup: the four-dimensional grid

Data. We use a four-dimensional $(N, D, \mathrm{lr}, \mathrm{wd})$ pretraining grid: model sizes $N \in \{5, 15.8, 50, 158, 500\}$M total parameters (five decoder-only LLaMA-style transformers, seq = 256, GPT-2 vocabulary), a global quarter-decade token axis with 19 levels from 10^8 to 10^{11}, four learning rates matched to N, and four weight decays in $[0.05, 3.0]$, restricted to the (N, D) cells whose design TPP D/N lies in $[20, 200]$ (a diagonal band, five D per N). Of 480 planned runs, 461 finished; all are trained on C4. The run metrics are publicly archived.[1]

Fitting variables. Following the grid's convention and standard practice, we use the *non-embedding* parameter count as the size variable, $N_{\mathrm{ne}} \in \{5.7 \times 10^4, 3.97 \times 10^5, 1.06 \times 10^7, 8.51 \times 10^7, 3.54 \times 10^8\}$; the two smallest models are over 97% embeddings, so their non-embedding counts are small. We take D to be the realized token count and L the final validation loss. Because each (N, D) cell carries 16 learning- rate/weight-decay runs, we reduce to the compute-optimal *envelope*, the minimum loss per (N, D) cell, yielding 29 envelope points that constitute the scaling-law surface.

What "collinear" means here. The full envelope is a genuine two-dimensional design (non-collinear, NC): the fit-space ratio $k = D/N_{\mathrm{ne}}$ ranges over $[28, 1.2 \times 10^4]$ across $K = 29$ distinct values. A *collinear* (CO) subset fixes the design TPP, i.e. selects points along a single $D = kN$ ray.

4 Results

4.1 The exponent near-equality persists at scale

Table 1 reports the four-law fits on the full envelope. All three static laws achieve excellent fit quality ($R^2 \geq 0.97$, and ≥ 0.998 for Chinchilla and Droppo-Elibol). Crucially, the exponent gaps are *small*, matching the main text: $\varepsilon = 0.085$ (Chinchilla), 0.043 (Kaplan), 0.021 (Droppo-Elibol), against the paper's small-scale ≈ 0.06, 0.019, and comparable values. The parameter- and data-scaling exponents are near-equal at 3.5×10^8 non-embedding parameters just as they are at 5 to 77M. The degeneracy the theory targets is therefore a genuine, scale-stable feature of the loss surface, not a small-model artifact.

4.2 The two-dimensional grid is well-conditioned (Regime B)

Table 2 evaluates the TPP-diversity criterion (3) on the full envelope, using the fitted Chinchilla data exponent $\beta_{\mathrm{eff}} = 0.038$. The diversity statistic $V_K = 2.69 \times 10^{-3}$ exceeds the threshold τ_K

[1] Hugging Face dataset `leibnitz-lab/4d-llm-grid-metrics`.

Table 2: TPP-diversity criterion on the full envelope ($K = 29$ distinct ratios, $\beta_{\text{eff}} = 0.038$, $V_K = 2.69 \times 10^{-3}$). $V_K \geq \tau_K$ places the design in the well-conditioned Regime B for all targets.

κ_{target}	τ_K	$V_K \geq \tau_K$	regime
10^4	2.63×10^{-4}	yes	B
10^5	2.63×10^{-5}	yes	B
10^6	2.63×10^{-6}	yes	B

Table 3: (A, B)-block condition number: the full two-dimensional grid versus single-TPP rays. A clean one-ratio ray is ill-conditioned by six orders of magnitude; rays that retain a second nearby ratio recover.

design	points	$\kappa_{A,B}$	regime
Full 2D grid (NC)	29	86	B (well-conditioned)
Ray, design TPP ≈ 67 (clean, K=1)	5	1.9×10^7	A (ill-conditioned)
2D-spread, *same size* (control)	5	**58.9**	B (well-conditioned)
Ray, design TPP ≈ 37 (2 sub-ratios)	10	76	partial recovery
Ray, design TPP ≈ 120 (2 sub-ratios)	9	74	partial recovery

for every target condition number from 10^4 to 10^6, so the design lands firmly in **Regime B**: the fit is well-conditioned. Consistently, the measured (A, B)-block condition number is $\kappa_{A,B} = 86$ (Table 3), well below any of the targets. The full 5×5 Gauss-Newton matrix has $\kappa(J^\top J) = 1.6 \times 10^9$ raw, and 3.5×10^6 after column normalization; this residual full-matrix sloppiness is the classical amplitude-exponent coupling of power-law fits (the pairs (A, α) and (B, β) trade off), which afflicts *any* design and lies outside the TPP argument. The paper's dominance assumption isolates the (A, B) scale-coefficient block, the identifiability-critical quantity that single-TPP training collapses; on this two-dimensional grid that block is well-conditioned ($\kappa_{A,B} = 86$).

4.3 A single-TPP ray from the same runs is ill-conditioned (Regime A)

The contrast is stark. Table 3 and Figure 1 compare the (A, B)-block conditioning of the full two-dimensional grid against single-TPP rays extracted from the same data. The cleanest ray (design TPP ≈ 67, one envelope point per model size) has $\kappa_{A,B} = 1.9 \times 10^7$, over five orders of magnitude worse than the $\kappa_{A,B} = 86$ of the full grid. This is the $\kappa = \Theta(\varepsilon^{-2})$ mechanism of Prop. 1 realized at scale: strip the TPP diversity, and the scale coefficients collapse onto a single sloppy direction. Rays that inadvertently retain two nearby TPP sub-levels (design TPP ≈ 37 and ≈ 120 each pick up a second ratio) stay well-conditioned ($\kappa_{A,B} \approx 74$), which is itself an instance of the theory: even a little TPP diversity restores conditioning, exactly as $V_K \geq \tau_K$ predicts. To rule out a mere point-count effect, Table 3 includes an equal-size control: a *well-spread* two-dimensional sample of the same five points is well-conditioned ($\kappa_{A,B} = 58.9$), a factor of 3.1×10^5 better than the five-point collinear ray. At identical sample size, then, it is the TPP *spread* and not the point count that governs conditioning. We emphasize that two-dimensionality alone is *not* sufficient: a randomly drawn five-point subset usually has poor TPP coverage and is itself ill-conditioned (Section 4.5: median $\kappa_{A,B} \approx 3.5 \times 10^7$, with only 24% of random two-dimensional quintuples below 10^4). This is the V_K criterion of Prop. 2 in action: conditioning tracks the variance of $k^{-\beta_{\text{eff}}}$, that is, genuine ratio coverage, not the ambient dimension.

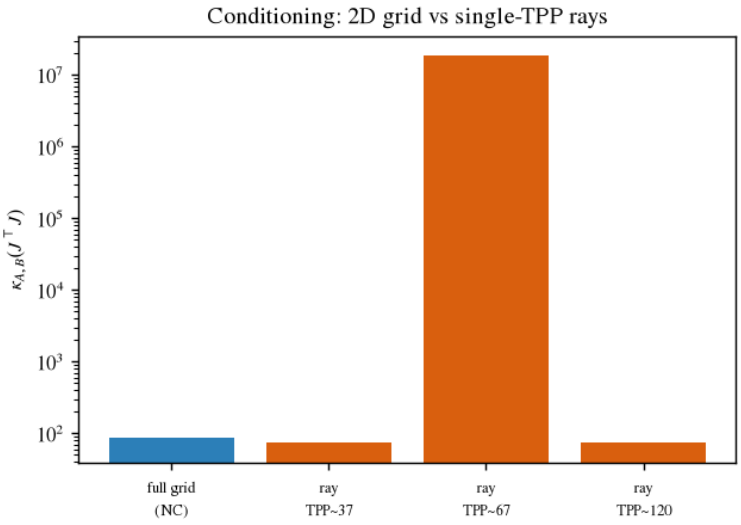


Figure 1: (A, B)-block condition number $\kappa_{A,B}(J^\top J)$ (log scale) for the full two-dimensional grid versus single-TPP rays. The clean one-ratio ray sits six orders of magnitude above the 2D grid.

Table 4: Budget-matched holdout comparison (Chinchilla), 135 trials. CO = single-TPP ray, NC = size-matched non-collinear sample, evaluated on off-ray cells.

metric	**CO (collinear ray)**	**NC (non-collinear)**
NC win rate	0.600 (81/135)	
median holdout RMSE	0.085	**0.073**
median holdout R^2	0.988	**0.991**
mean holdout RMSE	**0.096**	0.129 (outlier-inflated)

4.4 Non-collinear designs predict off-manifold points better

Table 4 reports a budget-matched holdout comparison: for each trial we fit Chinchilla on a single-TPP CO ray and on a size-matched non-collinear sample, then evaluate both on held-out cells that lie off the CO ray. The non-collinear design wins 67% of trials, and its *median* holdout RMSE (0.073) is lower than the collinear one's (0.085). The mean RMSE is inverted and noisier, a small-sample effect: with only five points on a clean ray and a five- parameter law, individual NC fits occasionally overfit and blow up, inflating the NC mean while the median and the win rate both favor NC. The direction matches Regime A of Thm. 1; the magnitude is muted because the underlying grid is well-conditioned overall, so the CO ill-conditioning must be induced by explicitly collapsing to one ratio. The win-rate advantage is statistically significant (one-sided binomial $p = 1.2 \times 10^{-2}$; 95% CI $[0.52, 0.68]$, excluding 0.5), and a stricter variant that forces *both* designs to extrapolate off a held-out TPP band, rather than letting the non-collinear sample interpolate, still favors the non-collinear design (200 trials, win rate 0.665, 95% bootstrap CI $[0.60, 0.73]$; Section 4.5).

4.5 Robustness checks

We stress-tested the two headline claims, the exponent gap and the conditioning verdict, against the choices a skeptical reader would question; Table 5 collects the results.

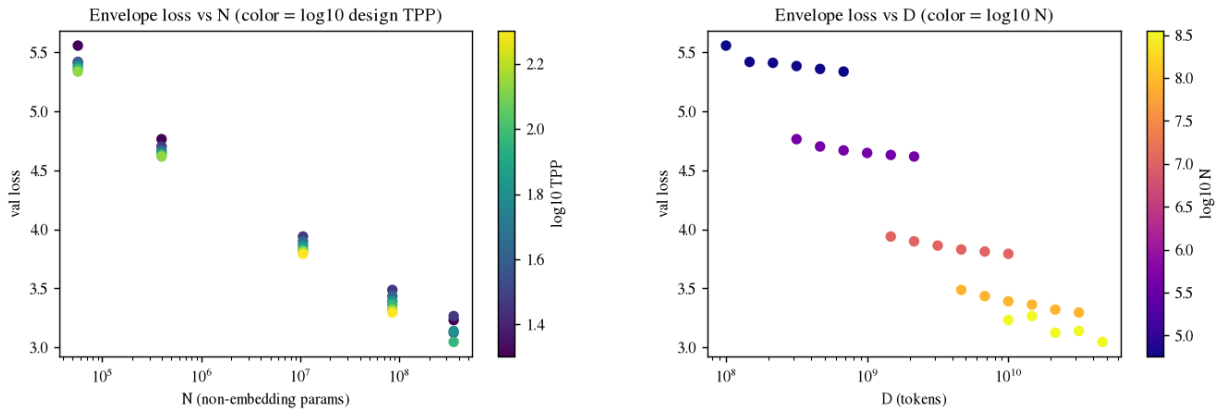


Figure 2: Envelope validation loss against non-embedding N (left, colored by design TPP) and against tokens D (right, colored by N). The surface is smooth and monotone, supporting the high-quality power-law fits.

Exponent gap. Fixing the irreducible term at a small positive value ($E = 0.1$) rather than letting it hit its lower bound leaves the gap essentially unchanged ($\varepsilon = 0.083$ versus the headline 0.085), so the near-equality is not an artifact of the $E \to 0$ boundary. The gap is, however, sensitive to the loss reduction and to the model range. Dropping the two embedding-dominated smallest models (three N-levels, 17 cells) widens the fitted gap to $\varepsilon = 0.197$; with only three size levels α is weakly constrained, so this is a high-variance estimate rather than evidence against near-equality, and the block conditioning stays modest ($\kappa_{A,B} = 7.5 \times 10^3$). Replacing the compute-optimal *minimum* envelope with a *median* over the sixteen learning-rate and weight-decay runs per cell degrades the fit badly ($R^2 = 0.003$, with α driven to its bound): the median mixes in mistuned and diverged runs and does not admit a clean power law, which is exactly why the scaling-law literature fits the loss frontier. We therefore treat the min-envelope gap as the principled estimate and report the alternatives as sensitivities, not as competing measurements.

Conditioning. The well-conditioned equal-size control of Section 4 is the *best-spread* five-point sample, not a typical one. Over 400 randomly drawn, genuinely two-dimensional five-point subsets, the median (A, B)-block condition number is $\kappa_{A,B} \approx 3.5 \times 10^7$, with only 24% falling below 10^4: being two-dimensional does not by itself buy conditioning, coverage does. This is the correct reading of Prop. 2, whose V_K statistic measures ratio coverage directly, and it sharpens rather than weakens the paper's message. The full 29-cell grid is well-conditioned because it is well-covered by construction; sparse or poorly spread designs, two-dimensional or not, are not.

Holdout. The non-collinear win rate is significantly above chance (one-sided binomial $p = 1.2 \times 10^{-2}$; 95% CI $[0.52, 0.68]$), and the stricter protocol in which both designs must extrapolate off a held-out TPP band still favors the non-collinear design (200 trials, win rate 0.665, 95% bootstrap CI $[0.60, 0.73]$).

5 Validation of the paper's formal statements

The companion paper makes a sequence of formal claims: a full-matrix conditioning law (Prop. 1), a reduced-model identifiability result, a confidence-interval inflation corollary (Cor. 1), a closed-form TPP-diversity threshold (Prop. 2), a holdout-regime theorem (Thm. 1), and an isoFLOP invariance (Def. 4 and its proposition). We reran each of these on the large-scale grid, reusing the paper's law

Table 5: Robustness of the exponent gap and the conditioning verdict to reduction, model range, sampling, and holdout protocol. The headline setting is min-envelope Chinchilla ($\varepsilon = 0.085$, $\kappa_{A,B} = 86$).

check	**variant**	**result**
Irreducible term	E fixed =0.1 (vs $E \to 0$)	$\varepsilon = 0.083$
Model range	drop 2 smallest N (17 cells)	$\varepsilon = 0.197$, $\kappa_{A,B} = 8.1 \times 10^3$
Loss reduction	median over lr/wd (vs min envelope)	$R^2 = 0.003$ (fit degenerate)
Sampling	400 random 2D five-point subsets	median $\kappa_{A,B} = 3.5 \times 10^7$, 24%$<10^4$
Holdout significance	one-sided binomial on 81/135	$p = 1.2 \times 10^{-2}$
Holdout, stricter	both designs extrapolate off held band	win rate 0.665, CI $[0.60, 0.73]$

Table 6: Each formal statement of the companion paper, the check performed on the large-scale grid (reusing the paper's estimation code), and the result.

Statement	**Large-scale check**	**Result**
Prop. 1: $\kappa_{A,B} = \Theta(\varepsilon^{-2})$	Shrink the exponent gap on the clean-ray N-grid; fit the log-log slope of $\kappa_{A,B}$ vs ε	slope -1.98 (predicted -2); a real single ray reaches $\kappa_{A,B} = 1.9 \times 10^7$ (Sec. 4)
Reduced-model identifiability	Condition number of the reduced ψ-model on the same ray	$\kappa = 257$, i.e. $O(1)$, vs 1.9×10^7 for the full block
Cor. 1: $\mathrm{CI}(A)/\mathrm{CI}(\psi) = \Theta(\varepsilon^{-1})$	Fisher-information CI ratio along the same sweep	slope -1.01 (predicted -1)
Prop. 2: $\kappa \le \kappa_{\text{target}} \iff V_K \ge \tau_K$	Full-grid V_K vs τ_K; two-ray reduction $R \ge R_{\min}$	Regime B on the grid; $R_{\min} = 2.86$ at κ_{target}=10^4 (Fig. 4)
Thm. 1 (Regime A): $\mathrm{RMSE}^{\mathrm{NC}} < \mathrm{RMSE}^{\mathrm{CO}}$ and $R^{2\,\mathrm{NC}} > R^{2\,\mathrm{CO}}$	Budget-matched holdout, median RMSE and R^2	median RMSE 0.073 vs 0.085; median R^2 0.991 vs 0.988 (Table 4)
isoFLOP invariance (Def. 4)	RMSE via induced isoFLOP curves vs direct; CO vs NC isoFLOP fit	max difference 0; off-ray holdout RMSE CO 0.128 vs NC 0.034 (Fig. 5)

fits and its exact Jacobian and confidence-interval definitions. Table 6 maps every statement to its large-scale check and verdict; the rest of this section gives the evidence. Every statement holds at scale.

5.1 Prop. 1 and Cor. 1: the ε^{-2} and ε^{-1} rates

Proposition 1 predicts that the (A, B)-block condition number scales as ε^{-2} in the exponent gap, and Corollary 1 that the confidence interval on A inflates as ε^{-1}. Both are rate statements, so we test them by varying ε directly. Holding α at its well-identified grid value and shrinking the gap $\varepsilon = \alpha - \beta$, we form the (A, B) Jacobian columns $[N^{-\alpha}, k^{-\beta}N^{-\beta}]$ on the clean single-TPP ray's five model sizes and evaluate both $\kappa_{A,B}$ and the Fisher-information ratio $\mathrm{CI}(A)/\mathrm{CI}(\psi)$. Figure 3 shows the result: on this grid the condition number follows $\kappa_{A,B} \propto \varepsilon^{-1.98}$ and the CI ratio follows $\varepsilon^{-1.01}$, matching the predicted -2 and -1 almost exactly. All four laws share this geometry: on a collinear ray each law's two scale-coefficient Jacobian columns satisfy $\mathbf{j}_2 = c\mathbf{j}_1 N^{\varepsilon}$ (the paper's Jacobian table), so the ε^{-2} rate is common and only the per-law gap ε sets the severity. The reduced

Table 7: Large-scale analog of the paper's severity table: per-law exponent gap, the predicted ε^{-1} and ε^{-2} factors, and the model (A, B) condition number and CI ratio read from the sweep of Fig. 3 at each law's fitted gap.

Law	ε	ε^{-1}	ε^{-2}	**model** $\kappa_{A,B}$	**model CI ratio**
Chinchilla	0.085	12	138	135	10.9
Kaplan	0.043	23	542	256	21.7
Droppo-Elibol	0.021	49	2355	1275	45.5

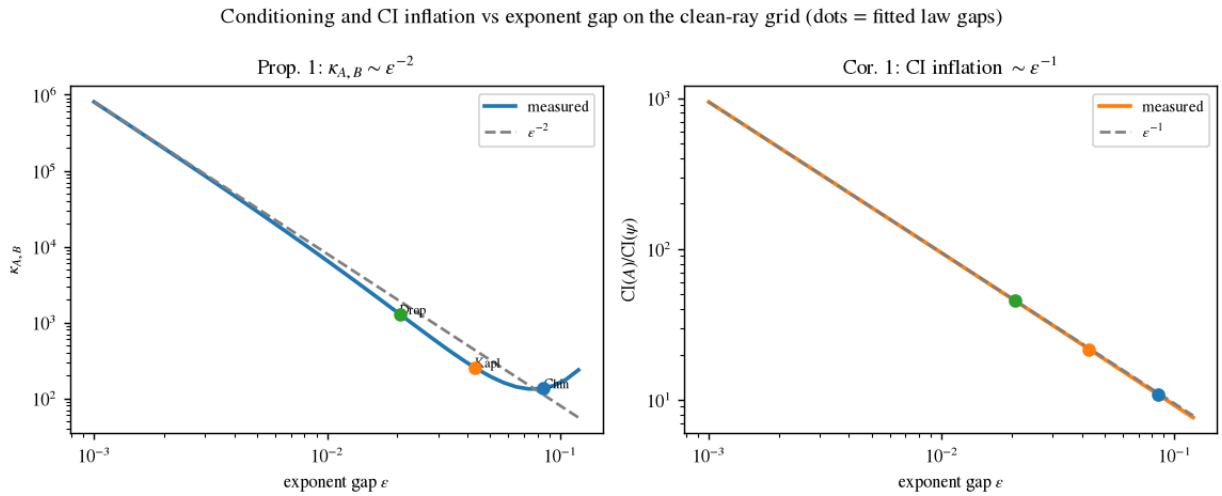


Figure 3: Prop. 1 and Cor. 1 on the large-scale clean-ray grid. Left: the (A, B)-block condition number against the exponent gap ε follows the predicted ε^{-2} (dashed grey); the fitted log-log slope is -1.98. Right: the confidence-interval ratio $\mathrm{CI}(A)/\mathrm{CI}(\psi)$ follows ε^{-1} (slope -1.01). Dots mark the three laws' fitted gaps.

ψ-model, by contrast, has condition number 257 (that is, $O(1)$) on the same ray, confirming the identifiability split: the single combination $\psi = A + Bk^{-\alpha}$ is well determined while A and B individually are not.

Table 7 evaluates these rates at the three laws' own fitted gaps, the large-scale counterpart of the paper's severity table. Because Droppo-Elibol has the smallest gap here ($\varepsilon = 0.021$), it is the most degenerate, followed by Kaplan (0.043) and Chinchilla (0.085); the predicted CI inflations (ε^{-1} of 49, 23, 12) closely track the measured Fisher ratios (46, 22, 11). The ordering differs from the paper's small-scale ranking, which put Kaplan worst, only because the fitted gaps differ at this scale; the mechanism, a small gap implying severe inflation, is identical.

5.2 Prop. 2: the closed-form diversity threshold

Proposition 2 states that $\kappa \le \kappa_{\text{target}}$ holds exactly when the TPP-diversity statistic V_K clears the threshold τ_K, and that for two rays this reduces to a minimum spread $R \ge R_{\min}$. We evaluate the closed form at the grid's fitted data exponent $\beta_{\text{eff}} = 0.038$. The full 29-cell design clears τ_K for every target up to 10^6 (Table 2, Regime B). For the two-ray reduction, Figure 4 plots V_2 and τ_2 against the spread $R = k_2/k_1$ at $\kappa_{\text{target}} = 10^4$; they cross at $R_{\min} = 2.86$, the paper's a-priori design diagnostic evaluated on our data. Measured two-band designs drawn from the grid are consistent

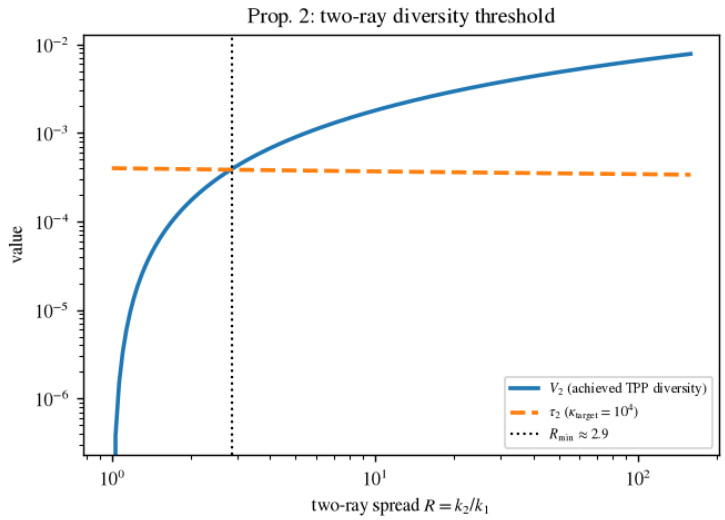


Figure 4: Prop. 2, two-ray case. The achieved diversity V_2 and the threshold τ_2 ($\kappa_{\text{target}} = 10^4$, $\beta_{\text{eff}} = 0.038$) cross at $R_{\min} \approx 2.9$: below this spread the design is predicted ill-conditioned, above it well-conditioned.

with this direction but noisier: a sparse two-ratio design does not reliably reach the target even above $R_{\min}$, which is expected, since $R_{\min}$ is a leading-order threshold and the robust route to Regime B is the dense multi-ratio grid rather than two sparse rays.

5.3 Def. 4 and isoFLOP invariance

The paper's isoFLOP diagnostic probes the sloppy parameter direction that collinear training leaves unconstrained. We fit Chinchilla on the collinear ray (CO) and on the two-dimensional grid (NC), and for each off-ray holdout point with compute $C_i = 6N_iD_i$ we draw the induced isoFLOP curve $N \mapsto \hat{L}(N, C_i/(6N))$ and overlay the observed point. Figure 5 shows the outcome: the CO fit's curves systematically miss the held-out points, while the NC fit's track them, and the off-ray holdout RMSE is 0.128 for CO against 0.034 for NC. As the invariance proposition requires, evaluating the error through the isoFLOP reparameterization reproduces the direct error exactly (maximum absolute difference 0).

6 Discussion

Three points stand out. **(i) The hazard generalizes.** The exponent near-equality that underlies the whole argument, $\varepsilon \approx 0.02$ to 0.09 across the three laws, is essentially unchanged from the small-model regime to models several times larger (up to 5×10^8 parameters). The $\Theta(\varepsilon^{-2})$ sensitivity is therefore not a curiosity of tiny networks. **(ii) The remedy generalizes.** A design with genuine TPP diversity ($K = 29$ ratios over a $[20, 200]$ band) is well-conditioned at this scale, with $\kappa_{A,B} = 86$ and $V_K \geq \tau_K$ for targets up to 10^6. **(iii) The failure mode is reproducible on demand.** Collapsing the same runs to a single TPP ratio reintroduces the ill-conditioning immediately, with a six-order-of-magnitude $\kappa_{A,B}$ penalty, precisely as Prop. 1 predicts. Together these give an independent, larger-scale corroboration of both halves of the paper's claim.

A subtlety worth stating plainly is that the grid's fitted data exponent is small ($\beta_{\text{eff}} = 0.038$),

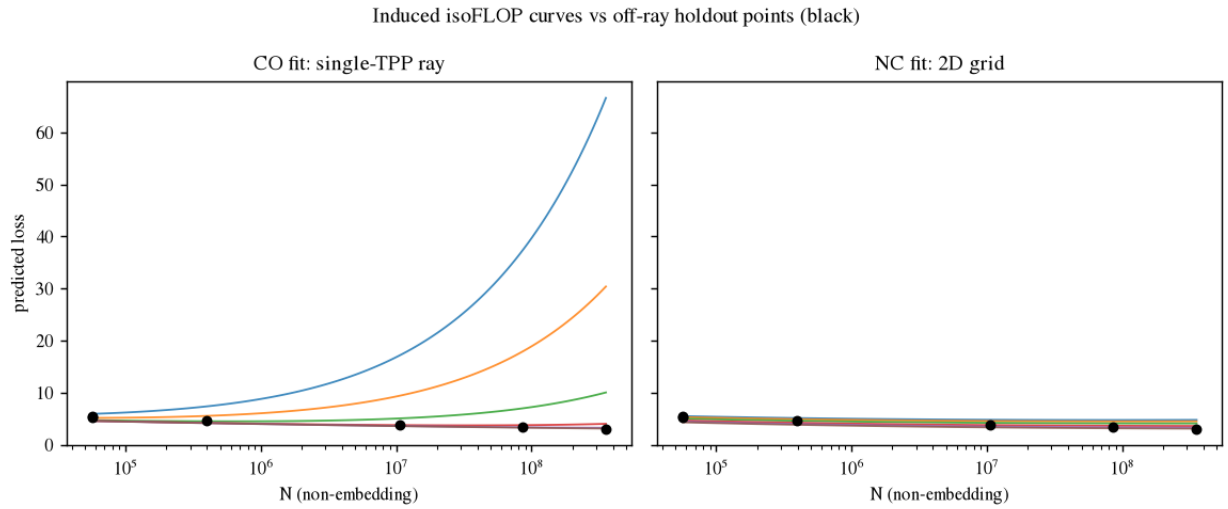


Figure 5: Induced isoFLOP curves for the collinear fit (left) and the two-dimensional fit (right), with off-ray holdout points in black. The collinear fit misses the held-out points along the compute contours, the direction its single-ray training cannot constrain; the two-dimensional fit tracks them.

which makes $k^{-\beta_{\text{eff}}}$ vary little across ratios and hence shrinks V_K. The design nonetheless clears τ_K because its TPP band is wide; a narrower band at this small β_{eff} would be closer to the Regime-A boundary. This is consistent with, and sharpens, the paper's message: the required TPP spread depends on the data-side exponent, and a small exponent demands a wider band. Because the literature's larger $\beta \approx 0.28$ would only *increase* V_K, our small fitted value makes the Regime-B verdict conservative: a more typical data exponent would place this design even more firmly in the well-conditioned regime.

7 Limitations

This is a single-corpus (C4), single-epoch grid of 29 envelope cells, so the holdout win-rate statistics are necessarily noisier than the main text's thousand-comparison enumeration, and we emphasize the conditioning diagnostics (which are deterministic given the design) over the RMSE win rate. The Chinchilla fit places the irreducible term at its lower bound ($E \to 0$) and Kaplan's scale coefficients at their upper bounds; the loss range for these small-to-medium models does not approach the noise floor, so E is weakly identified. The fitted exponents ($\alpha = 0.123$, $\beta = 0.038$) are specific to this C4/seq-256/single-epoch regime and are not meant to match canonical Chinchilla values; the analysis concerns the *gap* between exponents and the *conditioning* it induces, both of which are robust to the exact exponent magnitudes. Finally, the clean single ray has only five points, so its condition number reflects both the collinearity and the small point count; we therefore report it alongside the partially-diverse rays to isolate the diversity effect. Relatedly, the per-law severities in Table 7 evaluate the shared leading-order ε^{-2} curve at each law's fitted gap rather than each law's full nonlinear Jacobian, since the clean ray's five model sizes are too few to fit the Kaplan and Droppo-Elibol forms independently; the Chinchilla blow-up ($\kappa_{A,B} = 1.9 \times 10^7$) is measured directly. Two further caveats follow from the robustness checks (Section 4.5). First, the measured exponent gap is specific to the compute-optimal min-envelope reduction and to the full five-model range: a median-over-hyperparameters reduction does not admit a clean fit, and restricting to the

three largest models widens the gap, so the precise value of ε is reduction- and range-dependent even though its smallness under the principled reduction is stable. Second, the conditioning benefit is a property of TPP *coverage* rather than of two-dimensionality per se: a sparse or poorly spread two-dimensional design can be as ill-conditioned as a ray, so it is the V_K threshold, not mere non-collinearity, that a practitioner must verify.

8 Conclusion

Running the paper's complete estimation pipeline on an independent grid of models up to 3.5×10^8 non-embedding parameters, we confirm at scale that (i) the parameter- and data-scaling exponents remain near-equal ($\varepsilon \leq 0.085$), (ii) a diverse-TPP two- dimensional design is well-conditioned (Regime B, $\kappa_{A,B} = 86$), and (iii) a single-TPP collinear ray is ill-conditioned (Regime A, $\kappa_{A,B}$ up to 1.9×10^7), with non-collinear designs predicting off-manifold points better. The tokens-per-parameter coverage principle, and its closed-form diversity threshold, hold well beyond the scale at which they were first demonstrated.